# Week 6 Task 1: Introduction to Advanced Exploits

## Topics Covered

1. **Zero-Day Exploits** Overview of what zero-day exploits are, how they work, and their significance in cybersecurity.
2. **Understanding Buffer Overflow Attacks** Explanation of buffer overflow attacks and how they can be exploited.
3. **Basics of Reverse Engineering and Exploit Development** Introduction to reverse engineering and its role in exploit development.

## Objectives

- Understand the concept and execution of buffer overflow attacks.
- Learn the basics of exploit development through a tutorial.
- Set up a test environment using VulnHub, specifically "Earth: The Planet."

## Tasks Completed

## Definition: Zero day exploits

A zero-day exploit is a cyber attack that targets a software vulnerability that is unknown to the software vendor or developers. Since the developers don't yet know about it, there is no patch or fix, making it extremely dangerous.

"Zero-day" means the developers have had zero days to fix the flaw.

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## Why Are Zero-Day Exploits Dangerous?

- There's no defense (no patch, no antivirus signature)
- They can bypass security tools (firewalls, EDRs)
- Used by advanced hackers, APT groups, or nation-states

## 1. Research on Buffer Overflow Attacks

Buffer overflow attacks occur when a program attempts to write more data to a buffer than it can hold. This overflow can overwrite adjacent memory and is exploited by attackers to execute malicious code.

Key points:

- Attackers identify vulnerable code.
- Exploit payloads manipulate buffer memory.
- Control is gained by overwriting return addresses.

## 2. Tutorial on Exploit Development

Watched a tutorial covering:

- Setting up a development environment.
- Identifying vulnerabilities in programs.
- Crafting and testing exploit payloads.

  Link : [https://www.youtube.com/watch?v=cfqhilpcrRg](https://www.youtube.com/watch?v=cfqhilpcrRg)

## 3. Setting Up Test Environment with VulnHub

The "Earth: The Planet" vulnerable machine from VulnHub was used for hands-on learning. This machine provided practical insights into:

- Discovering and analyzing vulnerabilities.
- Exploiting the identified weaknesses step by step.
- Refining exploit development skills in a safe, controlled setup.